Reshaping data

Warmup activity

Work on the warmup activity (on the course website) with a neighbor, then we will discuss as a class

Warmup

```
# A tibble: 260 × 38
   country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`
`1983` `1984`
   <chr>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
<dbl>  <dbl>
 1 Afghan…     NA     NA     NA     NA   4.99   NA       NA     NA
NA   NA
 2 Albania     NA     NA     NA     NA  NA      NA       NA     NA
NA   NA
 3 Algeria     NA     NA     NA     NA  NA      NA       NA     NA
NA   NA
 4 Andorra     NA     NA     NA     NA  NA      NA       NA     NA
NA   NA
```

Question: What does a row in this data represent?

Warmup

```
# A tibble: 260 × 38
   country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`
`1983` `1984`
   <chr>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
<dbl>  <dbl>
 1 Afghan…     NA     NA     NA     NA   4.99   NA        NA     NA
NA   NA
 2 Albania     NA     NA     NA     NA  NA      NA        NA     NA
NA   NA
 3 Algeria     NA     NA     NA     NA  NA      NA        NA     NA
NA   NA
 4 Andorra     NA     NA     NA     NA  NA      NA        NA     NA
NA   NA
```

Question: What does a row in this data represent?

Each row is one country

Warmup

```
# A tibble: 260 × 38
   country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`
`1983` `1984`
   <chr>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
<dbl>  <dbl>
 1 Afghan…     NA     NA     NA     NA   4.99   NA        NA     NA
NA   NA
 2 Albania     NA     NA     NA     NA  NA      NA        NA     NA
NA   NA
 3 Algeria     NA     NA     NA     NA  NA      NA        NA     NA
NA   NA
 4 Andorra     NA     NA     NA     NA  NA      NA        NA     NA
NA   NA
```

Can I use the data in this form to make a scatterplot or fit a linear model?

Warmup

```
# A tibble: 260 × 38
   country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`
`1983` `1984`
   <chr>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
<dbl>  <dbl>
 1 Afghan…     NA     NA     NA     NA   4.99   NA        NA     NA
NA    NA
 2 Albania     NA     NA     NA     NA  NA      NA        NA     NA
NA    NA
 3 Algeria     NA     NA     NA     NA  NA      NA        NA     NA
NA    NA
 4 Andorra     NA     NA     NA     NA  NA      NA        NA     NA
NA    NA
```

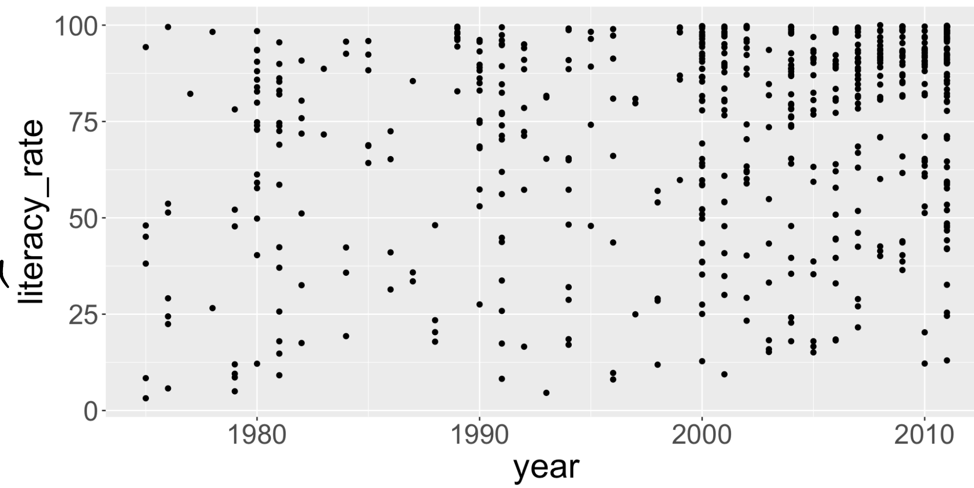
What do I want my data to look like instead?

country	year	literacy_rate
Afghanistan	1975	...
Afghanistan	1976	...
⋮	⋮	⋮

each row is now a country-year combination

What we want

	country	year	literacy_rate
1	Afghanistan	1979	4.987460
2	Afghanistan	2011	13.000000
3	Albania	2001	98.252274
4	Albania	2008	94.681814
5	Albania	2011	95.691480
6	Algeria	1987	35.839915
7	Algeria	2002	60.075082
8	Algeria	2006	63.918785
9	Angola	2001	54.194488
10	Angola	2011	58.608460
11	Anguilla	1984	95.714930
12	Antigua and Barbuda	2001	99.420000
13	Antigua and Barbuda	2011	99.420000
14	Argentina	1980	93.580894
15	Argentina	1991	96.041358
16	Argentina	2001	97.193411




```r
litF_long |>
  ggplot(aes(x = year,
             y = literacy_rate)) +
  geom_point()
```

plot(lit$year, lit$literacy_rate)

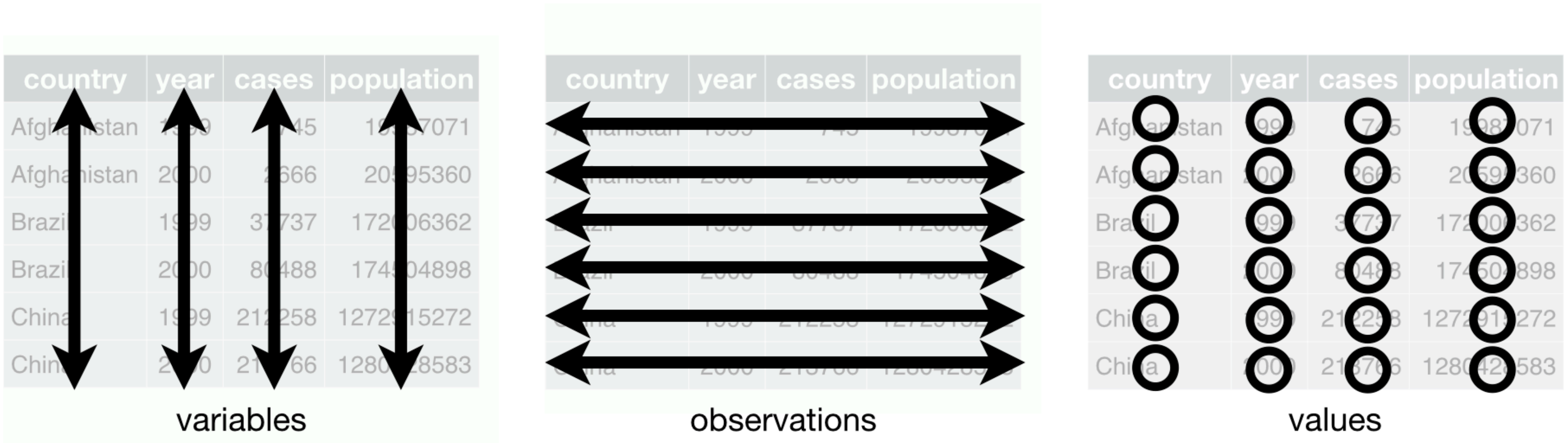
What we want

```
                  country year literacy_rate
1             Afghanistan 1979      4.987460
2             Afghanistan 2011     13.000000
3                 Albania 2001     98.252274
4                 Albania 2008     94.681814
5                 Albania 2011     95.691480
6                 Algeria 1987     35.839915
7                 Algeria 2002     60.075082
8                 Algeria 2006     63.918785
9                  Angola 2001     54.194488
10                 Angola 2011     58.608460
11               Anguilla 1984     95.714930
12      Antigua and Barbuda 2001     99.420000
13      Antigua and Barbuda 2011     99.420000
14              Argentina 1980     93.580894
15              Argentina 1991     96.041358
16              Argentina 2001     97.193411
```

```
  (Intercept)          year
-1323.2098674     0.6979597
```

```
lm(literacy_rate ~ year, data = litF_long)
```

Tidy data



(Image from *R for Data Science*)

Tidy data

Literacy data in tidy form:

```
# A tibble: 571 × 3
   country     year  literacy_rate
   <chr>       <chr>         <dbl>
 1 Afghanistan 1979           4.99
 2 Afghanistan 2011          13   
 3 Albania     2001          98.3 
 4 Albania     2008          94.7 
 5 Albania     2011          95.7 
 6 Algeria     1987          35.8 
 7 Algeria     2002          60.1 
 8 Algeria     2006          63.9 
 9 Angola      2001          54.2 
10 Angola      2011          58.6 
```

Another example

Data on three patients (A, B, C), with two blood pressure measurements (bp1 and bp2) per patient:

```
  id bp1 bp2
1  A 100 120
2  B 140 115
3  C 120 125
```

How might we want to reshape this data?

id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
	⋮	

Another example

Original data:

```
  id bp1 bp2
1  A 100 120
2  B 140 115
3  C 120 125
```

Reshaped data:

```
  id measurement value
1  A         bp1   100
2  A         bp2   120
3  B         bp1   140
4  B         bp2   115
5  C         bp1   120
6  C         bp2   125
```

Question: how do we do this reshaping in R?

Reshaping data: `pivot_longer`

(taking information in column names and moving into a single column)

id	bp1	bp2
A	100	120
B	140	115
C	120	125

id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

```
df |>
  pivot_longer(
    cols = bp1:bp2,
    names_to = "measurement",
    values_to = "value"
  )
```

← columns to pivot

← create a new column to store old column names

← create a new column to store old column values

(Image and example from *R for Data Science*)

pivot_longer

```
# A tibble: 260 × 38
   country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`
`1983` `1984`
   <chr>     <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
<dbl>  <dbl>
 1 Afghan…      NA     NA     NA     NA   4.99  NA        NA     NA
NA    NA
 2 Albania      NA     NA     NA     NA  NA     NA        NA     NA
NA    NA
 3 Algeria      NA     NA     NA     NA  NA     NA        NA     NA
NA    NA
 4 Andorra      NA     NA     NA     NA  NA     NA        NA     NA
NA    NA
```

```
litF |>
  pivot_longer(
    cols = ...,         ← everything except country
    names_to = ...,     ← year
    values_to = ...     ← literacy rate
  )
```

pivot_longer

```r
litF |>
  pivot_longer(
    cols = -country,
    names_to = "year",
    values_to = "literacy_rate"
  )
```

everything <u>except</u> country

```
# A tibble: 9,620 × 3
   country     year  literacy_rate
   <chr>       <chr>         <dbl>
 1 Afghanistan 1975         NA
 2 Afghanistan 1976         NA
 3 Afghanistan 1977         NA
 4 Afghanistan 1978         NA
 5 Afghanistan 1979          4.99
 6 Afghanistan 1980         NA
 7 Afghanistan 1981         NA
 8 Afghanistan 1982         NA
 9 Afghanistan 1983         NA
10 Afghanistan 1984         NA
```

pivot_longer

```
litF |>
  pivot_longer(
    cols = -country,
    names_to = "year",
    values_to = "literacy_rate"
  ) |>
  drop_na()  ← remove rows w/ NAs
```

```
# A tibble: 571 × 3
   country     year  literacy_rate
   <chr>       <chr>         <dbl>
 1 Afghanistan 1979           4.99
 2 Afghanistan 2011          13
 3 Albania     2001          98.3
 4 Albania     2008          94.7
 5 Albania     2011          95.7
 6 Algeria     1987          35.8
 7 Algeria     2002          60.1
 8 Algeria     2006          63.9
 9 Angola      2001          54.2
10 Angola      2011          58.6
```

pivot_longer

```
litF |>
  pivot_longer(
    cols = -country,
    names_to = "year",
    values_to = "literacy_rate",
    values_drop_na = T
  )
```

← drop rows with NA in value column (in literacy_rate)

```
# A tibble: 571 × 3
   country     year  literacy_rate
   <chr>       <chr>         <dbl>
 1 Afghanistan 1979           4.99
 2 Afghanistan 2011          13
 3 Albania     2001          98.3
 4 Albania     2008          94.7
 5 Albania     2011          95.7
 6 Algeria     1987          35.8
 7 Algeria     2002          60.1
 8 Algeria     2006          63.9
 9 Angola      2001          54.2
10 Angola      2011          58.6
```

Example 2

Now consider the following table:

```
ex_df
```

```
  id x_1 x_2 y_1 y_2
1  1   3   5   0   2
2  2   1   8   1   7
3  3   4   9   2   9
```

What will the following code return?

```
ex_df |>
  pivot_longer(cols = -id,
               names_to = "group_obs",
               values_to = "value")
```

everything except id

id	grup-obs	value
1	x_1	3
1	x_2	5
1	y_1	0
1	y_2	2
⋮	⋮	⋮

Example 2

Original data:

```r
ex_df
```

```
  id x_1 x_2 y_1 y_2
1  1   3   5   0   2
2  2   1   8   1   7
3  3   4   9   2   9
```

Reshaped data:

```r
ex_df |>
  pivot_longer(cols = -id,
               names_to = "group_obs",
               values_to = "value")
```

```
# A tibble: 12 × 3
      id group_obs value
   <dbl> <chr>     <dbl>
 1     1 x_1           3
 2     1 x_2           5
 3     1 y_1           0
 4     1 y_2           2
 5     2 x_1           1
 6     2 x_2           8
 7     2 y_1           1
 8     2 y_2           7
 9     3 x_1           4
10     3 x_2           9
11     3 y_1           2
12     3 y_2           9
```

Example 2

Original data:

```r
ex_df
```

```
  id x_1 x_2 y_1 y_2
1  1   3   5   0   2
2  2   1   8   1   7
3  3   4   9   2   9
```

x_1 → x 1

x_2 → x 2

y_1 → y 1

y_2 → y 2

Reshaped data:

```r
ex_df |>
  pivot_longer(cols = -id,
               names_to = c("group", "obs"),
               names_sep = "_",
               values_to = "value")
```

split name by _

```
# A tibble: 12 × 4
      id group obs   value
   <dbl> <chr> <chr> <dbl>
 1     1 x     1         3
 2     1 x     2         5
 3     1 y     1         0
 4     1 y     2         2
 5     2 x     1         1
 6     2 x     2         8
 7     2 y     1         1
 8     2 y     2         7
 9     3 x     1         4
10     3 x     2         9
11     3 y     1         2
12     3 y     2         9
```

Example 3

Consider the following example data:

bp_1 = blood pressure, stage 1
bp_2 = " " stage 2
hr_1 = heart , stage 1

```
  id bp_1 bp_2 hr_1 hr_2
1  1  100  120   60   77
2  2  120  115   75   81
3  3  125  130   80   93
```

What if we want the data to look like this:

```
# A tibble: 12 × 4
      id measurement stage value
   <dbl> <chr>       <chr> <dbl>
 1     1 bp          1       100
 2     1 bp          2       120
 3     1 hr          1        60
 4     1 hr          2        77
 5     2 bp          1       120
 6     2 bp          2       115
 7     2 hr          1        75
 8     2 hr          2        81
 9     3 bp          1       125
```

Example 3

```
df_3
```

```
  id bp_1 bp_2 hr_1 hr_2
1  1  100  120   60   77
2  2  120  115   75   81
3  3  125  130   80   93
```

```
df_3 |>
  pivot_longer(cols = -id,
               names_to = c("measurement", "stage"),
               names_sep = "_",
               values_to = "value")
```

separate names of original columns by _

```
# A tibble: 12 × 4
     id measurement stage value
  <dbl> <chr>       <chr> <dbl>
1     1 bp          1       100
2     1 bp          2       120
3     1 hr          1        60
4     1 hr          2        77
5     2 bp          1       120
6     2 bp          2       115
```

```
   7      2 hr           1         75
   8      2 hr           2         81
   9      3 bp           1        125
  10      3 bp           2        130
```

Example 3

```r
df_3 |>
  pivot_longer(cols = -id,
               names_to = c("measurement", "stage"),
               names_sep = "_",
               values_to = "value")
```

Step 1: Pivot

```
# A tibble: 6 × 3
     id measurement value
  <dbl> <chr>       <dbl>
1     1 bp_1          100
2     1 bp_2          120
3     1 hr_1           60
4     1 hr_2           77
5     2 bp_1          120
6     2 bp_2          115
```

Step 2: Separate columns

```
# A tibble: 6 × 4
     id measurement stage value
  <dbl> <chr>       <chr> <dbl>
1     1 bp          1       100
2     1 bp          2       120
3     1 hr          1        60
4     1 hr          2        77
5     2 bp          1       120
6     2 bp          2       115
```

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)

For next time, read:

- Chapter 5 in *R for Data Science* (2nd ed.)